📄 📊 Employee Attrition & Performance Analysis – Report

1. Introduction

Employee attrition is a major challenge for organizations as it increases operational costs and affects productivity. This project focuses on analyzing employee data to identify key factors that influence employee turnover.

The analysis is performed using data visualization and machine learning techniques to gain meaningful insights and support decision-making.

2. Objective

- To analyze employee attrition patterns
- To identify key factors influencing employee turnover
- To create an interactive dashboard using Power BI
- To build a machine learning model for attrition prediction

3. Dataset

The dataset used is the **IBM HR Analytics Employee Attrition Dataset**, which includes employee-related features such as:

- Age
- Department
- Job Role
- Monthly Income
- Job Satisfaction
- Work-Life Balance
- Years at Company
- Attrition (Yes/No)

4. Workflow

1. Data Collection
2. Data Cleaning & Preprocessing
3. Exploratory Data Analysis (EDA)
4. Data Visualization
5. Machine Learning Model
6. Dashboard Development
7. Insights & Conclusion

5. Data Cleaning

- Missing values were handled using forward fill method
- Duplicate records were identified and removed
- Data types were corrected for accurate analysis

6. Data Analysis & Visualization

Various visualizations were created to analyze employee attrition:

- Attrition by Department
- Salary vs Attrition (Average Income)
- Age Group vs Attrition
- Job Satisfaction vs Attrition
- Work-Life Balance vs Attrition
- Years at Company vs Attrition
- Correlation Heatmap

These visualizations helped in identifying patterns and relationships between variables.

7. Machine Learning Model

A **Decision Tree Classifier** was used to predict employee attrition.

- Input Features:
    - Age
    - Monthly Income
    - Job Satisfaction
    - Work-Life Balance
    - Years at Company

- The model achieved good accuracy in predicting whether an employee will leave or stay.

This shows that salary, satisfaction, and experience are key factors affecting attrition.

8. Power BI Dashboard

An interactive dashboard was created using Power BI with:

- KPI Cards:
  - Total Employees
  - Employees Left
  - Attrition Rate
- Visuals:
  - Attrition by Department
  - Salary vs Attrition
  - Age Group Analysis
  - Job Satisfaction
  - Work-Life Balance
  - Years at Company
- Features:
  - Slicers (Filters)
  - Drill-through functionality

9. Test Cases

- Filter Check: Dashboard updates based on selected department
- Calculation Check: Attrition rate calculation verified
- Drill-through: Detailed employee data accessible

10. Key Insights

- Sales department has the highest attrition
- Employees with lower salaries are more likely to leave
- Younger employees show higher turnover
- Poor work-life balance increases attrition

11. Conclusion

Employee attrition is influenced by multiple factors such as salary, job satisfaction, and work-life balance.

Improving these areas can help organizations reduce employee turnover and improve retention.

12. Recommendations

- Improve salary and compensation structure
- Enhance work-life balance policies
- Increase employee engagement
- Reduce workload in high-pressure departments

🔗 13. Project Link

GitHub Repository:
https://github.com/abhijeetbarai/Internship_project1